Mastering AI & Data Science Interviews
80 Interview Questions for BCA & MCA Freshers

Part 1

Machine Learning • Deep Learning • LLM • Generative AI • RAG • MLOps

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Module 1 - Machine Learning Fundamentals

Q1. What is Machine Learning?

Interview Question

What is Machine Learning?

Answer (Interview Ready)

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed.

Example: Netflix recommends movies based on your watch history.

Applications: Recommendation Systems, Fraud Detection, Healthcare, Self-driving Cars

Follow-up Questions

• Is ML a subset of AI? Yes.

• Can ML work without data? No.

Q2. Types of Machine Learning

Interview Question

Types of Machine Learning

Answer (Interview Ready)

There are four main types: Supervised, Unsupervised, Semi-Supervised and Reinforcement Learning.

Example: Customer Segmentation uses Unsupervised Learning.

Applications: Prediction, Clustering, Robotics

Follow-up Questions

• Which uses labeled data? Supervised.

• Which uses rewards? Reinforcement Learning.

Q3. Difference between Supervised, Unsupervised and Reinforcement Learning

Interview Question

Difference between Supervised, Unsupervised and Reinforcement Learning

Answer (Interview Ready)

Supervised learns from labeled data, Unsupervised finds hidden patterns, and Reinforcement learns through rewards and penalties.

Example: Spam Detection, Customer Segmentation, Self-driving Cars.

Applications: Classification, Clustering, Robotics

Follow-up Questions

• Which is most common? Supervised.

Q4. What is a Machine Learning Pipeline?

Interview Question

What is a Machine Learning Pipeline?

Answer (Interview Ready)

A Machine Learning Pipeline is the complete process of collecting data, preparing it, training a model, evaluating it, deploying it, and monitoring its performance.

Example: House Price Prediction.

Applications: Any ML Project

Follow-up Questions

• Why evaluate a model? To measure performance.

Q5. What are Overfitting and Underfitting?

Interview Question

What are Overfitting and Underfitting?

Answer (Interview Ready)

Overfitting means the model learns the training data too well. Underfitting means it fails to learn enough from the data.

Example: Student memorizing answers vs studying too little.

Applications: Model Optimization

Follow-up Questions

• High variance causes? Overfitting.

Q6. What are Bias and Variance?

Interview Question

What are Bias and Variance?

Answer (Interview Ready)

Bias occurs when a model is too simple. Variance occurs when a model is too complex and learns noise.

Example: Simple vs overly complex model.

Applications: Model Selection

Follow-up Questions

• High bias causes? Underfitting.

Q7. What is Feature Engineering?

Interview Question

What is Feature Engineering?

Answer (Interview Ready)

Feature Engineering is creating or selecting useful input features to improve model performance.

Example: Extracting month from a date.

Applications: Prediction Models

Follow-up Questions

• What is a feature? An input variable.

Q8. What is Feature Scaling?

Interview Question

What is Feature Scaling?

Answer (Interview Ready)

Feature Scaling brings numerical features to a similar range.

Example: Scaling Age and Salary.

Applications: KNN, SVM, Neural Networks

Follow-up Questions

• Do Decision Trees need scaling? No.

Q9. What is Cross Validation?

Interview Question

What is Cross Validation?

Answer (Interview Ready)

Cross Validation checks model performance using different parts of the dataset.

Example: K-Fold Cross Validation.

Applications: Model Evaluation

Follow-up Questions

• Most common method? K-Fold.

Q10. What are Evaluation Metrics?

Interview Question

What are Evaluation Metrics?

Answer (Interview Ready)

Evaluation Metrics measure how well a machine learning model performs.

Example: Accuracy for spam detection.

Applications: Model Comparison

Follow-up Questions

• Name one classification metric: Accuracy.